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Visual Intelligence

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Benchmarking Drag⋆ for eye direction transformation and beyond

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Abstract

Eye direction plays a crucial role in determining the quality of photographs containing human faces. Images where subjects look in inconsistent directions are often perceived as low-quality and discarded. While state-of-the-art deep generative models such as DragGAN, DragDiffusion, and DragonDiffusion (collectively referred to as Drag⋆) offer potential solutions for eye direction transformation, their effectiveness for this specific task remains unexplored. In this work, we systematically investigate the capability of Drag ⋆ for eye direction transformation. Our initial experiments reveal that these models in their original form cannot effectively perform this task. To address this limitation, we construct a specialized dataset (i.e., eye multi-direction dataset (EMDD)) and establish a comprehensive benchmark for evaluating methods through fine-tuning on our curated data. Our analysis demonstrates that fine-tuned models achieve satisfactory results when the angular difference between the directions of the source and target eyes is small. However, we observe significant performance degradation when large directional changes are necessary. Through detailed investigation, we uncover the underlying causes of these limitations and provide insights into the models' failure modes. To overcome these challenges, we propose the *edge-localized point selector* and *zero-latent source region replacement*, which can alleviate the identified limitations. Experimental results demonstrate that our approach achieves substantial performance improvements for eye direction transformation, particularly in scenarios involving large angular changes.

Keywords: Eye direction transformation, DragGAN, DragDiffusion, DragonDiffusion

1 Introduction

In the past few years, image editing techniques have been advanced rapidly [1–3], with facial image editing emerging as a particularly important area due to its relevance in applications such as digital media, identity protection, and virtual communication [4–6]. Of all the aspects of facial editing, manipulating the eyes is one of the most challenging tasks, as this region of the face is highly sensitive and even subtle changes can have a significant impact on the perceived realism [7, 8]. In particular, eye direction plays a critical role across various fields. For instance, eye tracking and gaze estimation [9–11] are utilized in automotive safety research to study drivers' attention and concentration in different traffic conditions [12, 13]; in psychology, these techniques are used to investigate attention, cognitive load, and social behaviors [14]. In human-computer interaction, eye-based inputs facilitate interaction with the surrounding environment and machinery [15].

Recent progress in generative models has provided powerful tools for eye direction transformation. Notably, DragGAN [16], DragDiffusion [17], and DragonDiffusion [18]—collectively referred to as Drag⋆—enable fine-grained image edits by specifying source and target points in the latent space. These methods have demonstrated impressive performance on a wide range of editing tasks, including object relocation, geometric shape manipulation, and localized attribute modification.

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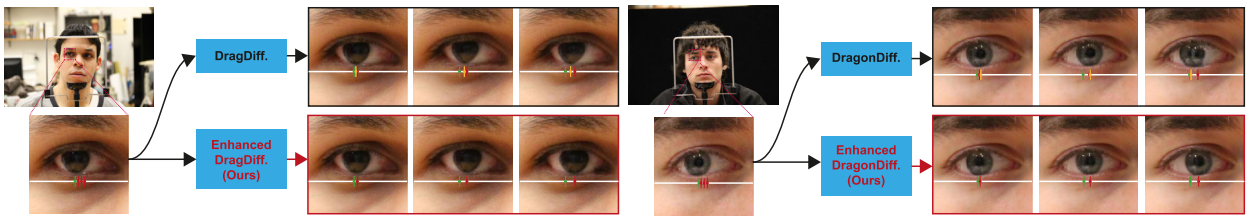

Figure 1 Examples of eye direction transformation via the original DragDiffusion and DragonDiffusion and our enhanced versions. We show the source eye directions with green points and the target eye directions with red points, and the results of original DragDiffusion and DragonDiffusion with yellow points. Clearly, the original versions fail when the angular difference between the source and the target is large, whereas our enhanced versions succeed. (Source from the Columbia Gaze dataset [19])

Despite their notable strengths, Drag⋆ methods still exhibit clear limitations when applied to specialized tasks. In particular, they often struggle even with seemingly simple eye direction transformations. To examine this issue systematically, we investigate the behavior of Drag⋆ in the context of gaze editing. Our preliminary experiments show that, in their original form, these models cannot consistently achieve reliable modifications of eye direction, especially when the target gaze involves large angular shifts. As illustrated in Fig. 1, when the transfer angle is large, the editing quality deteriorates substantially. The methods often fail to redirect the gaze accurately to the target region or produce severe image distortions. This observation highlights a fundamental challenge: while effective for general-purpose image editing, existing latent manipulation frameworks are not well-aligned with the structural and perceptual constraints of the human eye, underscoring the need for more task-specific solutions.

To facilitate rigorous evaluation, we introduce the eye multi-direction dataset (EMDD), a curated dataset containing images of eyes with diverse gaze directions based on the Columbia Gaze dataset [19]. Using EMDD, we establish a comprehensive benchmark for assessing gaze-editing methods. Experiments show that fine-tuned versions of Drag⋆ achieve satisfactory results for small angular changes, but performance degrades significantly under larger gaze shifts. Further analysis identifies potential causes of these limitations, including misalignment in latent representations and insufficient modelling of long-range directional transformation tasks.

Building on these insights, we propose two solutions: an edge-localized point selector for point-based methods (e.g. DragGAN, DragDiffusion) and a zero-latent source region replacement for score-based methods (e.g. DragonDiffusion). Together, these strategies substantially improve performance, particularly in scenarios involving large angular changes. As shown in Fig. 1, applying our improvements significantly enhances the quality of gaze transfer, resulting in more accurate redirection and fewer visual artifacts. Extensive experiments on EMDD demonstrate that our approach outperforms existing Drag⋆-based solutions and achieves state-of-the-art results.

The overview of this work is shown as Fig. 2. In summary, the main contributions of this work are as follows:

1) We construct the eye multi-direction dataset (EMDD) and establish a dedicated benchmark, highlighting limitations of existing Drag⋆ methods for eye direction transformation.
2) We analyze the causes of these limitations, providing insights into why general-purpose latent editing models struggle with large-angle gaze changes.
3) We propose an edge-localized point selector and zero-latent source region replacement to address these challenges and improve performance on eye direction transformation tasks.

2 Related works

2.1 Facial editing

Facial attribute editing focuses primarily on modifying specific facial attributes in an image while preserving the overall identity and realism of the face. Early approaches, such as CycleGAN [22], utilized cycle-consistent adversarial networks to achieve unpaired image-to-image translation, thereby enabling facial attribute manipulation. StarGAN [23] introduced the concept of multi-domain facial attribute editing, using a single unified network to perform transformations across multiple attribute domains. By incorporating domain labels to guide the generation process, this method achieves diverse and precise attribute modifications across different domains. Approaches based on local feature transfer [24, 25] modify the latent representations of specific local regions in the image, inducing desired changes in those areas while keeping the rest of the image unchanged. As a result, facial attributes can be edited by replacing or altering local image regions. DeFLOCNet [26] facilitates fine-grained editing by allowing direct control over low-level image features—such as color, texture, and shape. This method also supports user-guided inputs (e.g., sketches or color cues) to generate facial images that align with the user's expectations. In con-

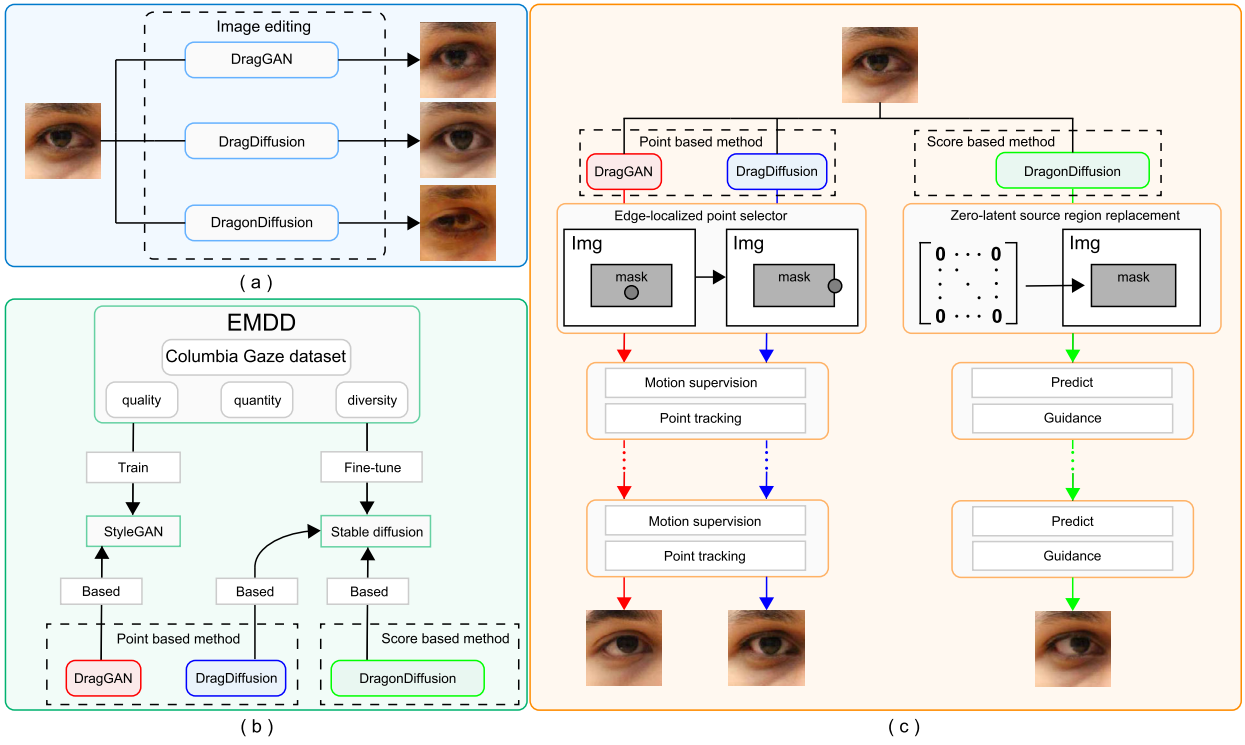

Figure 2 Overview of this work. (a) Preliminary study for Drag★ methods; (b) the eye-multi-direction dataset we construct for eye direction transformation; (c) our improvements to the Drag★ methods used for the eye direction transformation task. Image editing approaches include DragGAN [16], DragDiffusion [17], and DragonDiffusion [18]. We used StyleGAN [20] and Stable diffusion [21] to build our eye multi-direction dataset (EMDD)

trast to prior editing approaches, our method offers significant advantages, including precise spatial manipulation, effective single-target editing, and well-preserved consistency in surrounding areas.

2.2 Gaze redirection

Gaze redirection holds significant application value in the fields of computer vision and image processing. Early gaze redirection techniques [27] primarily relied on geometric models, which simulated eye rotation by constructing a geometric representation of the eyeball to achieve redirected gaze. Novel view synthesis methods [22, 28–30] generate the desired redirection by rendering the scene containing the subject's face from a specified viewpoint, simulating the effect of looking into the camera. These approaches require acquiring depth maps of the face and then applying 3D transformations to synthesize new images with redirected gaze. However, they often depend on specialized hardware to obtain depth information and modify the entire scene, which limits their practicality. Monocular gaze synthesis also aims to alter the gaze direction within the eye region. For instance, DeepWarp [31] employs a neural network to predict a dense flow field, which warps the input image to the desired gaze direction. Unlike these gaze redirection techniques, our technique avoids the reliance on heavy hardware, explicit geometric modeling, or optical flow prediction by directly generating the redirected image from 2D data, thereby greatly simplifying the pipeline.

2.3 Latent space and diffusion-based editing

Beyond facial attribute manipulation with GAN-based models, some works have focused on more general image editing approaches. A representative direction is latent space inversion and manipulation in StyleGAN [20]. Images can be inverted into the latent space of StyleGAN, after which semantic attributes are modified by shifting latent vectors [32–34]. This approach enables high-quality reconstructions and precise attribute control within a well-structured latent representation. Building on this idea, DragGAN [16] introduces an interactive point-to-point dragging scheme. Operating in the StyleGAN latent space, it allows users to select source and target points on an image and then optimizes the latent code such that the corresponding semantic features are aligned. This enables intuitive and fine-grained geometric manipulation.

In parallel, the success of diffusion models [21] has motivated a new class of editing methods [35, 36], where edits

are performed in the latent space of a variational autoencoder (VAE). To address this, recent work such as DragDiffusion [17] and DragonDiffusion [18] has extended the interactive dragging mechanism to diffusion models. By optimizing the diffusion latent space, these models combine the high-quality and diverse generative capacity of diffusion models with fine-grained, user-controllable manipulation.

Compared to general tasks, eye direction transformation tasks require the editing process to retain much more detail, including the skin around the eyes, the basic shape of the eyeball and eye socket, and even the background.

3 Problem and preliminary study

3.1 Problem formulation

Let $I \in \mathbb{R}^{H \times W}$ denote a facial image, and $E \subset \Omega$ represent the eye region, where $\Omega = \{1, \ldots, H\} \times \{1, \ldots, W\}$ is the set of all pixel coordinates in image I. The goal of *eye direction transformation* is to generate a modified image I', in which the gaze of the eye shifts toward a target direction, θ, while the rest of the image remains unchanged:

$$Diff(gaze(E), gaze(E')) = \theta, \tag{1}$$

$$s.t. \quad \forall (x, y) \in \Omega \backslash E, I'(x, y) \approx I(x, y), \tag{2}$$

where $E' \subset \Omega$ denotes the transformed eye region in modified image I', and $gaze(\cdot)$ estimates the gaze direction.

In latent-space generative frameworks Drag⋆, the image I is encoded into a latent representation L. The task is then to find a latent perturbation ΔL, which produces the gaze change:

$$I' = G(L + \Delta L) \Rightarrow Diff(gaze(E), gaze(E')) = \theta, \tag{3}$$

where G denotes the generator mapping from latent space to image space.

3.2 Preliminary study

In our preliminary study, we employed the dlib library to detect faces in the input images and extracted two separate 512×512 patches, one centred on the left eye and one centred on the right eye. Existing editing methods were then applied to these patches to make fine-grained modifications to the pupil position. The results, however, were unsatisfactory, as shown in Fig. 2(a). The models often failed to preserve fine-grained details, and the generated gaze directions were inaccurate. While pre-trained models such as StyleGAN2 and Stable Diffusion 1.5 demonstrated reasonable performance in their original training domains, they performed poorly in our specific task. These observations highlight a substantial domain mismatch between existing pre-trained models and the requirements of eye direction editing, indicating that off-the-shelf methods are insufficient. To address this gap, we propose the construction of a specialized dataset, the eye multi-direction dataset (EMDD), tailored to enable systematic evaluation and adaptation of gaze editing methods.

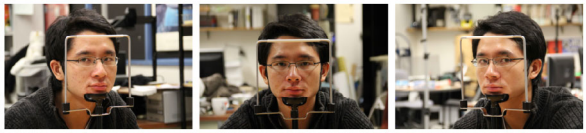

Figure 3 The examples of varying lighting and poses

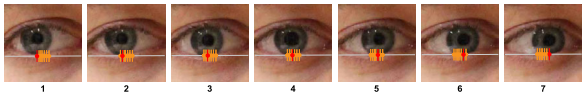

Figure 4 The seven yaw angles for eye directions. The center of the gaze is red, and the points are spaced 5° apart on the eyeball

4 Eye Multi-Direction Dataset (EMDD)

From our previous attempts, it becomes evident that existing methods and pre-trained models are insufficient for accomplishing our specified task, primarily due to a significant domain shift between the available data and the data required for our target application. Therefore, establishing a dedicated dataset tailored to our task is crucial.

4.1 Dataset construction

When establishing the dataset, we took into account the quantity, quality, and diversity of the data sources. On one hand, the selected data need to cover a wide range of gaze directions and possess as high a resolution as possible to retain more detailed information. On the other hand, it is also essential to include a variety of eye shapes, skin tones, iris colors, and other attributes to enhance diversity. Based on these criteria, we selected the Columbia Gaze dataset [19] as our data source. This is a high-resolution public dataset containing eye images of 56 subjects. As shown in Fig. 3, the images incorporate varied head poses and lighting conditions to reflect real-world scenarios. Each head pose is associated with 21 distinct gaze directions, formed by combinations of seven yaw angles $[-15°, -10°, -5°, 0°, 5°, 10°, 15°]$ (as shown in Fig. 4). We employed the dlib library to parse these images, detecting 68 facial landmarks to accurately localize the eye regions. These regions were then cropped and used to construct a dataset consisting of over 8,000 images at a resolution of 512×512 pixels.

5 Benchmarking Drag ⋆ on EMDD

To explore the ability to reveal the shortcomings of existing methods, we designed a series of experiments in this section.

5.1 Methods under evaluation

To evaluate the effectiveness of generative models in eyeball editing, we employed three representative editing methods: DragGAN [16], DragDiffusion [17], and DragonDiffusion [18]. Given that these methods are widely used in eye editing area and we selected them as the target methods for our evaluation.

DragGAN: A point-based interactive dragging model built upon the StyleGAN2 architecture, which iteratively performs motion supervision and point tracking to progressively achieve the editing objective.

DragDiffusion: A point-based interactive manipulation model based on the Stable Diffusion network, which also requires motion supervision and point tracking steps.

DragonDiffusion: A score-based gradient-guided model built upon the Stable Diffusion framework. It achieves editing goals by applying guidance to different regions, steering the original predictions to guide the generation process toward the target.

5.2 Experiment settings

5.2.1 Preparation

For all experiments in this section, the dataset was randomly split into a training set and a testing set in a 9: 1 ratio. The training set was used directly to train the generative models employed by the eye editing methods (see Sect. 5.2.2). The testing set was processed such that each individual's ocular images include multiple horizontal gaze angles, with the central forward gaze designated as the reference, and was used to evaluate the eye editing methods (see Sect. 5.2.3).

5.2.2 Generative model training

The generative models serve as the foundation for our eye-editing framework. We train a StyleGAN2 network on the training set, which serves as the backbone for the *DragGAN*-based editing pipeline. Separately, we fine-tune the *VAE* and *U-Net* components of the Stable Diffusion 1.5 architecture on the same training set. This model provides the generative basis for the *DragDiffusion* and *DragonDiffusion* pipelines.

5.2.3 Testing set preprocessing

In contrast to the training set, where the models were trained directly on the original images, the testing set required additional preparation to meet the input specifications of the editing models. Specifically, to provide the models with input in their expected format, namely, the definition of source–target correspondences and the selection of mask regions, we manually annotated a set of landmark points P and corresponding mask regions M for each test image in advance. These annotations were stored and subsequently used during the editing process.

Furthermore, due to the highly aligned nature of our dataset, it can be reasonably assumed that, within each test group, non-pupil regions (e.g., periocular texture and eye socket position) exhibit strong consistency, with the primary variation occurring in pupil location. This characteristic enables the use of a single reference image to manipulate the pupil direction to match that of other target images. The resulting edited images can subsequently be evaluated against their corresponding real images through both quantitative and qualitative comparisons.

Table 1 Target and achieved editing angles for the three methods. The table presents the desired angles (left) alongside the actual angles achieved by each method (right) for the corresponding editing tasks. The values in parentheses indicate the deviation from the target angle. Tasks where the deviation exceeds 3° are highlighted in gray. We observe that the deviation from the target angle is significantly greater when the target angle exceeds 20°

Target	DragGAN	DragDiffusion	DragonDiffusion
5°	5°(0°)	5°(0°)	5°(0°)
10°	10°(0°)	10°(0°)	10°(0°)
15°	13°(2°)	13°(2°)	14°(1°)
20°	14°(6°)	14°(6°)	14°(6°)
25°	15°(10°)	15°(10°)	15°(10°)
30°	15°(15°)	15°(15°)	16°(14°)

5.3 Experiment design

To better explore the performance of existing methods, a two-stage (coarse-to-fine) experimental design was adopted. In the coarse stage, a preliminary evaluation of the editing efficacy of the three methods was performed. The success of an edit was measured by the displacement of the edited pixels. This displacement was converted into a rotation angle (rounded to the nearest integer) and was used as the primary metric.

As revealed by the results in Table 1, the deviation from the target angle is significantly greater when the target angle exceeds 20°. In particular, the methods' performances tend to degrade as the gaze transformation angle increases, indicating a sensitivity to larger shifts. This observation suggests that gaze transformation behavior may vary significantly across angular thresholds, warranting a more detailed analysis. To quantitatively investigate this further, we designed a two-phase experiment. Specifically, we designed two experiments focusing on different gaze transformation scenarios.

1) Small eye angular shift experiments. A small eye angular shift means that the angular difference between the directions of the source and target eyes is small, and we define a small shift as being less than 20°. In this experiment, we use the central angle image E_4 of each individual as the reference. Images with pupils gazing toward the edges E_1 to E_3 and E_5 to E_7 were edited to align with the central direction (e.g., $E_1 \rightarrow E_4$, $E_7 \rightarrow E_4$), while the central image E_4 was also edited to simulate extreme left and right gaze directions (e.g., $E_4 \rightarrow E_1$, $E_4 \rightarrow E_7$).

2) Large eye angular shift experiments. In contrast, a large angular shift of the eye indicates a significant difference in the directions of the source and target eyes. In this work, a large shift is defined as greater than 20°. Experiments are performed on all angle pairs with differences greater than 20° among the seven yaw angles (e.g., $E_1 \rightleftharpoons E_6$, $E_2 \rightleftharpoons E_7$), to evaluate the resulting displacement performance.

By employing these experiments, we can assess editing performance across varying angular ranges and generate symmetrically paired results for each operation, in order to support a more thorough and robust evaluation.

5.3.1 Editing procedure

Among the three editing models employed, both DragGAN and DragDiffusion operate based on a mechanism of moving supervision and point tracking, whereas DragonDiffusion utilizes a score-based gradient guidance.

For DragGAN and DragDiffusion, each editing task was configured as follows: the initial image for editing, E_o, was selected; the pupil center in E_o was defined as the source point P_1, and the pupil center in the target image E_t as the target point P_2. The entire orbital region was designated as the editable mask area. These parameters were then input into the respective editing model, yielding an edited output image E_e.

For DragonDiffusion, the selection of source and target points followed the same configuration. The mask region was divided into two parts: the entire pupil of E_o was designated as the target mask, mask_{src}, while the sclera region was selected as the reference mask, mask_{ref}. These parameters were then provided to the model to generate the edited image E_e.

5.4 Evaluation metrics

To evaluate the effectiveness of eye-editing methods, we conducted a quantitative comparison of the edited image E_e and the predetermined ground truth target image E_t using the following three metrics:

1) Mean squared error (MSE): It captures pixel-level discrepancies, providing a straightforward measure of pixel-wise error.
2) Structural similarity index measure (SSIM): It evaluates structural and contrast similarity, aligning with human perceptual quality and demonstrating robustness to variations in luminance and contrast.
3) Learned perceptual image patch similarity (LPIPS): It measures high-level semantic differences by leveraging deep network features, effectively capturing perceptual variations in texture, shape, and other semantic attributes.

Together, these metrics allow us to measure low-level fidelity, mid-level structural consistency, and high-level semantic similarity, enabling a holistic evaluation of both the accuracy and effectiveness of the eye editing.

5.5 Experiment results

The detailed results are presented in Table 2 and Table 3. For each row of editing metrics, we average the results across test data groups under the same angular editing task as the final results.

5.5.1 Small eye angular shift experiments

Based on the results presented in Table 2, the following conclusions can be drawn:

1) In the small-angle alignment experiment configuration, all three editing models demonstrated relatively stable performance in the task of eyeball editing with small-angle variations. Moreover, the direction of editing did not significantly affect performance, as symmetric editing pairs exhibited negligible differences in results.
2) The performance of the three models varied. In general, the diffusion-based models achieved superior results. This phenomenon may be attributed to the fact that the reconstruction method used with the StyleGAN network yields a poorer reconstruction quality compared to the approach employed in diffusion models. If the image reconstructed by a GAN-based method differs from the ground truth, subsequent editing is unlikely to effectively preserve details, which explains the performance gap between DragGAN and the other two models.
3) It is easy to observe that DragonDiffusion achieved notably higher scores in SSIM. This may be due to its distinct implementation mechanism. Unlike point-driven editing approaches, DragonDiffusion relies on score-based gradient guidance, which may better preserve structural details of the original region compared to point manipulation, thereby resulting in higher structural similarity.

5.5.2 Large eye angular shift experiments

Based on the results presented in Table 3, the following conclusions can be drawn:

1) Unlike the relatively stable performance of the three models in small eye angular shift experiments, which involved small-angle variations, their performance in large eye angular shift experiments showed a noticeable correlation with the angular range of the editing task. Specifically, as the angular disparity in the editing task increased, the performance of the models degraded. This trend was particularly evident in the two diffusion-based models: from the $E_1 \rightarrow E_5$ combination (20°) to the $E_1 \rightarrow E_7$ combination (30°), the LPIPS and MSE scores progressively increased, while the SSIM scores decreased accordingly.
2) Compared with the editing angle, the editing direction has little impact on the results, especially in terms of LPIPS and SSIM, whereas the MSE metric exhibits a much larger variation.

Table 2 Experiment 1: Small eye angular shift experiments. The results of the three metrics for the three models are shown for 12 actions. The Best result for each metric under each action is underlined and bold

	DragGAN			DragDiffusion			DragonDiffusion		
Action	LPIPS ↓	SSIM ↑	MSE↓	LPIPS ↓	SSIM ↑	MSE↓	LPIPS ↓	SSIM ↑	MSE↓
$E_1 \rightarrow E_4$	0.378	0.701	826.83	**0.250**	0.733	**302.51**	0.252	**0.734**	309.84
$E_4 \rightarrow E_1$	0.384	0.706	935.24	**0.252**	0.728	**311.47**	0.257	**0.732**	324.68
$E_2 \rightarrow E_4$	0.369	0.704	938.75	0.230	0.760	**204.07**	**0.219**	**0.762**	204.88
$E_4 \rightarrow E_2$	0.380	0.708	870.86	**0.220**	0.759	**208.30**	0.223	**0.760**	218.64
$E_3 \rightarrow E_4$	0.373	0.709	786.30	**0.231**	0.764	**200.90**	0.233	**0.766**	204.80
$E_4 \rightarrow E_3$	0.370	0.722	833.59	0.225	0.765	202.48	**0.225**	**0.770**	**193.76**
$E_5 \rightarrow E_4$	0.374	0.703	856.44	**0.226**	0.747	**221.88**	0.228	**0.749**	225.83
$E_4 \rightarrow E_5$	0.381	0.707	886.44	**0.228**	**0.748**	**226.88**	0.230	0.748	227.86
$E_6 \rightarrow E_4$	0.388	0.692	1059.83	**0.230**	0.752	**242.52**	0.235	**0.755**	242.96
$E_4 \rightarrow E_6$	0.381	0.707	899.27	0.236	0.750	250.97	**0.234**	**0.757**	**228.28**
$E_7 \rightarrow E_4$	0.399	0.685	1167.31	**0.234**	0.753	240.317	0.235	**0.756**	**237.77**
$E_4 \rightarrow E_7$	0.380	0.706	892.16	0.241	0.744	267.62	**0.234**	**0.756**	**239.94**

Table 3 Experiment 2: Large eye angular shift experiments. The results of the three metrics for the three models are shown for 12 actions

	DragGAN			DragDiffusion			DragonDiffusion		
Action	LPIPS ↓	SSIM ↑	MSE↓	LPIPS ↓	SSIM ↑	MSE↓	LPIPS ↓	SSIM ↑	MSE↓
$E_1 \rightarrow E_7$	0.378	0.701	870.79	0.262	0.722	390.45	**0.254**	**0.729**	**361.79**
$E_7 \rightarrow E_1$	0.403	0.683	1290.77	**0.253**	0.729	**339.51**	0.255	**0.732**	347.98
$E_1 \rightarrow E_6$	0.379	0.703	880.03	0.242	0.736	309.80	**0.234**	**0.746**	**255.24**
$E_6 \rightarrow E_1$	0.393	0.692	1176.38	0.235	0.738	296.49	**0.232**	**0.747**	**248.31**
$E_1 \rightarrow E_5$	0.378	0.702	855.87	0.219	0.764	198.06	**0.211**	**0.772**	**154.46**
$E_5 \rightarrow E_1$	0.376	0.705	891.99	**0.215**	0.762	201.26	0.217	**0.769**	**178.66**
$E_2 \rightarrow E_7$	0.377	0.695	1039.60	0.236	0.740	288.78	**0.223**	**0.755**	**233.15**
$E_7 \rightarrow E_2$	0.407	0.679	1231.71	0.230	0.748	246.35	**0.229**	**0.752**	**245.95**
$E_2 \rightarrow E_6$	0.377	0.697	987.25	0.233	0.736	**296.20**	**0.227**	**0.745**	304.35
$E_6 \rightarrow E_2$	0.392	0.686	1106.75	0.230	0.739	287.80	**0.228**	**0.748**	**259.40**
$E_3 \rightarrow E_7$	0.377	0.703	849.35	0.237	0.753	251.50	**0.229**	**0.766**	**207.39**
$E_7 \rightarrow E_3$	0.394	0.692	1177.45	0.224	0.765	213.41	**0.220**	**0.769**	**195.72**

Based on the analysis and experimental results presented above, it can be concluded that the three existing editing methods, combined with the proposed dataset, have successfully addressed the fundamental challenge of enabling eyeball editing from scratch. Furthermore, diffusion-based editing models were observed to perform relatively well for adjustments involving small angular variations. However, all three methods exhibited limited effectiveness in handling large angular disparities. During practical implementation, we noted that in large angular editing tasks, all approaches manifested a phenomenon akin to insufficient dragging, where the models failed to achieve the desired displacement magnitude as specified. This indicates that current methods have certain limitations. To enhance the performance of these methods in eyeball editing tasks, the following section will explore and analyze the underlying issues affecting large angular editing performance and propose targeted improvements accordingly.

6 Beyond Drag ★

In this section, we focus on the challenge of insufficient dragging discussed in Sect. 5.5.

6.1 Cause analysis

To facilitate a more fundamental analysis of the problem, we are reclassifying the three models using criteria different from the previous GAN- and Diffusion-based categorizations. Instead, we distinguish them based on their inherent editing mechanisms. Accordingly, the three models are divided into two categories:

1) Point interaction-based editing models: This includes DragGAN and DragDiffusion, which rely on interactive point dragging to achieve edits, as shown in Fig. 5.
2) Score-based gradient guidance editing model: Represented by DragonDiffusion, which utilizes score-based gradient guidance for predictive editing, as shown in Fig. 6.

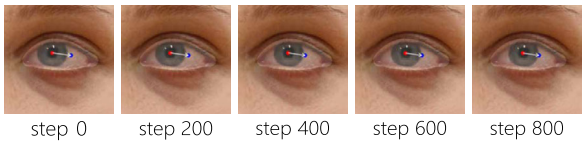

Figure 5 Point interaction-based editing models. The bottom indicates the editing iteration count (each iteration consists of one set of motion supervision and point tracking operations). The white area represents the editable region, which is constrained to limit the editing scope. The red point denotes the source point p_d updated by point tracking at the current step, and the blue point indicates the target point p_t

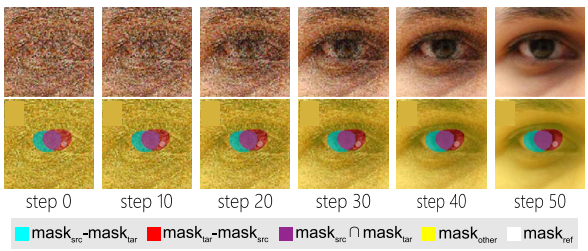

Figure 6 Score-based gradient guidance editing model. The first row displays the evolution of the generated images corresponding to the latent variables during the denoising process. The second row shows schematic diagrams of the initially input mask variable positions on the corresponding generated images. Yellow mask represents the non-editable region $\mathbf{mask}_{other}$, red mask indicates the target region $\mathbf{mask}_{tar}$, purple mask corresponds to the overlapping area between the source mask $\mathbf{mask}_{src}$ and $\mathbf{mask}_{tar}$, cyan mask denotes the non-overlapping area of $\mathbf{mask}_{src}$ and $\mathbf{mask}_{tar}$, and white mask designates the reference region $\mathbf{mask}_{ref}$

Although both point-based interactive dragging and score-based gradient-guided editing exhibit similar symptoms in large angular tasks, namely insufficient displacement and oscillatory behavior, the underlying causes differ due to their distinct mechanisms.

6.1.1 Point-based interactive dragging methods

For dragging-based methods, ΔL is guided by a driving point (or start point) p_d and the target location point p_t as

$$\Delta L = \text{argmin}_{\Delta L} \mathcal{L}_{\text{point}}(G(L + \Delta L), \{p_d, p_t\}), \tag{4}$$

where $\mathcal{L}_{\text{point}}$ measures the enforcement of consistency between the moved points in latent space and the given target location p_t.

The cause lies in the selection of the pupil center as the driving point. The feature representations of pixels around the pupil center are highly homogeneous, leading to two key issues: (1) Vanishing gradients in motion supervision: The similarity among neighboring pixels reduces the effectiveness of L_1 loss, resulting in negligible updates during backpropagation. (2) Instability in point tracking: When many neighbouring pixels share nearly identical features, the tracking step is prone to mismatching, causing the driving point to oscillate locally rather than progress toward the target direction. Together, these factors explain the observed hesitation and oscillatory movement that occurs during large angular dragging.

To solve this problem, we propose an edge-localized point selector to enable more stable and effective long-range editing. (See Sect. 6.2)

6.1.2 Score-based gradient guidance methods

Existing score-based, gradient-guided editing methods follow a similar workflow. A source point P_1 and a target point P_2 define the editing direction $\boldsymbol{D} = P_1 - P_2$, while a source mask mask_{src} specifies the editable region and a reference mask mask_{ref} provides contextual guidance. The input image is mapped to a latent variable L, which is partially shifted along $\boldsymbol{D}$ to obtain a modified latent $L' = L + \Delta L$ and a corresponding target mask mask_{tar}.

$$\Delta L = \text{argmin}_{\Delta L} \mathcal{L}_{\text{region}}(G(L + \Delta L), \text{mask}_{\text{tar}}, \theta), \tag{5}$$

where $\mathcal{L}_{\text{region}}$ measures the discrepancy between the current and target gaze directions θ within mask_{tar}.

In the practice of DragonDiffusion, before the optimization iteration starts, the update begins with

$$L \leftarrow L \cdot (1 - \text{mask}_{\text{tar}}) + L' \cdot \text{mask}_{\text{tar}}). \tag{6}$$

In this way, the features of the source region could be transferred to the target region via translational mapping.

During the denoising stage, there are two signals introduced: (1) an *editing guidance* G_e derived by comparing mask_{tar} and the original mask_{src} and (2) a *reference guidance* G_r obtained by comparing mask_{tar} and mask_{ref}. The two signals are combined with the U-Net prediction to refine the latent representation. However, in practice, the effective editing distance may not be sufficient.

Upon re-examining the intermediate images generated during the denoising process, we hypothesize that the limited editing displacement arises from the initial pre-processing step. This pre-processing is designed to pre-populate the target region with latent features of the object, so as to reduce the reliance on subsequent guidance. In practice, however, its effectiveness is limited: even with pre-positioned latents and additional guidance, the predictions are not consistently directed toward the desired content in the target region.

Moreover, after translating mask_{src}, we observed that the model often produces pupil predictions in between mask_{src} and mask_{tar}, rather than precisely within the target region. As a result, the achieved displacement is insufficient, and the editing performance further degrades as the required displacement distance increases.

Figure 7 Point-based editing with edge-localized point selector. To mitigate the issues of vanishing gradients in motion supervision and instability in point tracking, we propose a point selector that chooses the driving point based on the edge detection rather than the pupil center. The red point denotes the source point updated by point tracking at the current step, and the blue point indicates the target point. The driving point is selected as a local center of the edge region within the target pupil area

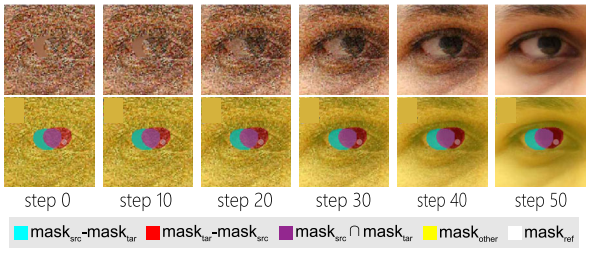

Figure 8 Score-based method with zero-latent source region replacement

To solve this problem, we propose a method that replaces the mask_{src} region with a zero latent L_{zero}. (See Sect. 6.3)

6.2 Edge-localized point selector for Drag ★

To mitigate the issues of vanishing gradients in motion supervision and instability in point tracking, we propose a point selector that chooses the driving point based on the edge detection rather than the pupil center. The key idea is that the points along the pupil-sclera boundary exhibit more distinctive local features than those near the center, which will bring two advantages:

1) Enhanced motion supervision: Stronger local feature contrasts produce larger gradients, improving the effectiveness of backpropagation for directional editing.
2) Improved point tracking: Distinctive features around edge points reduce the likelihood of mismatches, stabilizing the tracking of the driving point.

Formally, given an initial eye region E_{o}, we detect the pupil boundary using a standard edge detector (e.g., Canny [37]). The driving point p_{d} is then selected as a local center of the edge region within the target pupil area. Subsequent editing operations, including motion supervision and point tracking, are performed with respect to p_{d} instead of the original pupil center, as shown in Fig. 7.

Algorithm 1 Edge-localized point selector

Require: Initial image E_{o}, target pupil mask $mask_{\text{t}}$, edge detector EdgeDetector()
Ensure: Driving point p_{d}
  Function SelectEdgePoint($E_{\text{o}}, \text{mask}_{\text{t}}$, **EdgeDetector**)
    edges $\leftarrow$ *EdgeDetector*($E_{\text{o}} \odot \text{mask}_{\text{t}}$)
    candidate_points $\leftarrow \{(x, y) \mid \text{edges}(x, y) = 1\}$
    $p_{\text{d}} \leftarrow$ compute_local_center(candidate_points)
  return p_{d}

Algorithm 2 Zero-latent source region replacement

Require: Latent variable L, source mask mask_{src}, target mask mask_{tar}
Ensure: Edited latent variable ΔL
  Function RegionReplace($L, \text{mask}_{\text{src}}, \text{mask}_{\text{tar}}$)
    $L[\text{mask}_{\text{src}}] \leftarrow 0$
    $\Delta L \leftarrow \text{ModelGenerate}(L, \text{mask}_{\text{tar}})$
  return ΔL

This edge-based driving point selection significantly enhances the stability and effectiveness of long-range dragging-based edits, addressing the vanishing gradient and oscillation issues observed with center-based points. The corresponding pseudocode is presented in Algorithm 1.

6.3 Zero-latent source region replacement

To overcome the limitation of insufficient editing distance, we propose a simple strategy during the pre-processing of the latent variable L. Specifically, instead of retaining the image information in the mask_{src} region, we replace this region with a zero matrix. Note that, due to the varying distances of editing, the target mask may overlap with the original mask, and such overlapping regions are preserved rather than being filled with zeros.

This strategy has two benefits for editing. Firstly, the original editing process exhibited a tendency to generate the pupil between the mask_{tar} and mask_{src} regions rather than within the intended mask_{tar} area. By filling the mask_{src} region with a zero matrix during pre-processing, we disrupted the original data distribution, thereby breaking this inherent prediction bias. This could encourage the network to generate the pupil predominantly within the mask_{tar} region. Secondly, the zero-filled mask_{src} region promotes the generation of sclera-like content—characterized by whiter and structurally consistent features—in the area originally covered by mask_{src}. This aligns with the expected anatomical context and enhances the overall coherence of the editing, as shown in Fig. 8. The corresponding pseudocode is presented in Algorithm 2.

Table 4 Drag⋆ with our method in the large eye angular shift experiments. Better results are labelled in bold

	LPIPS ↓	SSIM ↑	MSE↓	LPIPS ↓	SSIM ↑	MSE↓	LPIPS ↓	SSIM ↑	MSE↓	LPIPS ↓	SSIM ↑	MSE↓
	$E_1 \rightarrow E_7$			$E_7 \rightarrow E_1$			$E_1 \rightarrow E_6$			$E_6 \rightarrow E_1$		
DragGAN	0.378	0.701	870.79	0.403	0.683	1290.77	0.379	0.703	880.01	0.393	0.692	1176.38
DragGAN+ours	**0.362**	**0.719**	**710.76**	**0.387**	**0.699**	**1132.69**	**0.367**	**0.715**	**755.00**	**0.384**	**0.700**	**1091.85**
DragDiffusion	0.262	0.722	390.45	0.253	0.729	339.51	0.242	0.736	309.80	0.235	0.738	296.49
DragDiffusion+ours	**0.243**	**0.736**	**294.09**	**0.244**	**0.734**	**305.42**	**0.232**	**0.745**	**246.56**	**0.224**	**0.750**	**232.83**
DragonDiffusion	0.254	0.729	361.79	0.255	0.732	347.98	0.234	0.746	255.24	**0.232**	0.747	248.31
DragonDiffusion+ours	**0.245**	**0.741**	**298.19**	**0.249**	**0.741**	**269.74**	**0.231**	**0.751**	**235.22**	0.233	**0.751**	**227.96**
	$E_1 \rightarrow E_5$			$E_5 \rightarrow E_1$			$E_2 \rightarrow E_7$			$E_7 \rightarrow E_2$		
DragGAN	0.378	0.702	855.87	0.376	0.705	891.99	0.377	0.695	1039.60	0.407	0.679	1231.71
DragGAN+ours	**0.365**	**0.716**	**743.49**	**0.366**	**0.714**	**810.10**	**0.366**	**0.707**	**908.91**	**0.395**	**0.688**	**1139.16**
DragDiffusion	0.219	0.764	198.06	0.215	0.762	201.26	0.236	0.740	288.78	0.230	0.748	246.35
DragDiffusion+ours	**0.213**	**0.768**	**167.88**	**0.210**	**0.767**	**181.88**	**0.225**	**0.750**	**240.69**	**0.220**	**0.758**	**211.88**
DragonDiffusion	**0.211**	**0.772**	**154.46**	0.217	0.769	178.66	**0.223**	0.755	233.15	0.229	0.752	245.95
DragonDiffusion+ours	0.214	0.771	166.23	**0.215**	**0.771**	**160.91**	0.224	**0.758**	**231.12**	**0.226**	**0.761**	**207.10**
	$E_2 \rightarrow E_6$			$E_6 \rightarrow E_2$			$E_3 \rightarrow E_7$			$E_7 \rightarrow E_3$		
DragGAN	0.377	0.697	987.25	0.392	0.686	1106.75	0.377	0.703	849.35	0.394	0.692	1177.45
DragGAN+ours	**0.366**	**0.710**	**897.22**	**0.381**	**0.699**	**1039.00**	**0.363**	**0.718**	**734.84**	**0.382**	**0.702**	**1098.06**
DragDiffusion	0.233	0.736	296.20	0.230	0.739	287.80	0.237	0.753	251.50	0.224	0.765	213.41
DragDiffusion+ours	**0.227**	**0.740**	**274.36**	**0.223**	**0.748**	**253.83**	**0.229**	**0.764**	**209.05**	**0.220**	**0.768**	**200.87**
DragonDiffusion	**0.226**	0.745	304.35	0.229	0.748	259.41	0.229	0.766	207.39	**0.220**	0.769	195.72
DragonDiffusion+ours	0.228	**0.747**	**286.70**	**0.228**	**0.753**	**243.13**	**0.228**	**0.772**	**187.64**	**0.220**	**0.773**	**183.55**

7 Validation and discussion

7.1 Settings and design

This section conducts experimental tests to evaluate the effectiveness of the previously proposed method, primarily examining the performance of the improved workflow when handling large angular editing tasks. With the exception of the adjusted workflow settings in Sect. 6.2 and Sect. 6.3, all other experimental configurations remain identical to those in large eye angular shift experiments of Sect. 5.3.

7.2 Discussion

Table 4 reports quantitative results for DragGAN, DragDiffusion, and DragonDiffusion, both with and without our improvements, across four large eye angular shift experiments. Across most of test cases, our method consistently improves all baselines: LPIPS decreases, SSIM increases, and MSE decreases, indicating more accurate and visually coherent edits. For example, DragGAN+ours reduces LPIPS by up to 0.013 and MSE by 107 compared to DragGAN, while DragDiffusion+ours and DragonDiffusion+ours achieve similar improvements. Even state-of-the-art DragonDiffusion benefits from our improvements, achieving lower LPIPS and higher SSIM across all tasks. Only in a single case (DragonDiffusion on $E_1 \rightarrow E_5$ experiment) did we observe a marginal decrease in performance, yet the overall trend strongly validates the effectiveness and robustness of our approach. These results demonstrate that our method effectively enhances both perceptual quality and reconstruction accuracy in long-range eye editing tasks.

Table 5 The counts of successful detections in the original and edited generated images

Generated images	Count
Origin	35
DragGAN+ours	35
DragDiffusion+ours	35
DragonDiffusion+ours	35

7.3 Further analysis

7.3.1 Perceptual evaluation

To better assess the impact of our strategy and the three state-of-the-art editing methods on the perceptual quality of generated eye regions, we trained a separate YOLOv8 model for perceptual detection of eyes. With the confidence threshold configured to 0.5, experimental results in Table 5 demonstrate that both the original images and the edited versions maintained perceptual consistency.

7.3.2 Runtime analysis

To better measure the computational overhead introduced by our method, we evaluated the computational time of three baseline methods both with and without the integration of our strategies. As shown in Fig. 9, the results indicate that our strategic improvement introduces negligible additional computational overhead. This observation is consistent with our theoretical design, as the operations

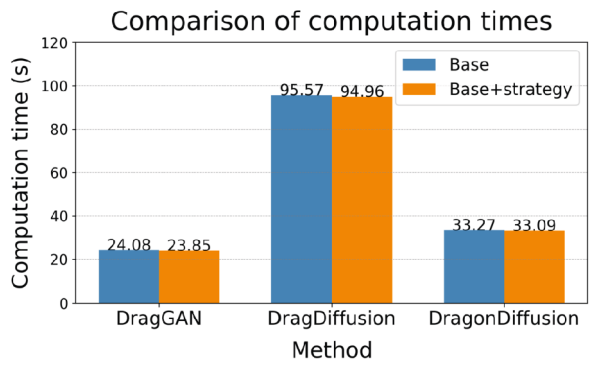

Figure 9 Average image editing time consumption

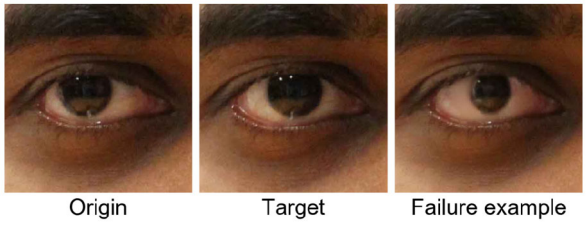

Figure 10 Failure case

involved in our strategies are of $O(1)$ complexity. Consequently, the image editing process with our strategies incurs almost no extra time compared to the original methods.

8 Limitations and future work

8.1 Failure discussion

Through our experimental results in Table 4, it can be observed that a small number of groups showed insignificant improvement, while very few groups even exhibited a slight decline in performance. This phenomenon is indeed related to the different strategies and methods employed. On the one hand, the second strategy involves more substantial intervention in image reconstruction, and in cases where the editing angle is not particularly large, the filled zero vectors may introduce certain interference to the reconstruction process—especially when there is excessive overlap between the original and target regions, as shown in Fig. 10. On the other hand, the three methods, namely DragGAN, DragDiffusion, and DragonDiffusion, all demonstrate varying levels of performance. It can be noted that DragonDiffusion almost consistently achieves the best baseline editing results. Therefore, it is understandable that our strategy is more effective at improving the performance of the other two methods than it is at improving DragonDiffusion's performance. Nevertheless, even though the original method already delivers strong performance, our strategy still enhances the editing effectiveness in the majority of groups and scenarios, and achieves more significant improvements, particularly under large-angle editing conditions.

8.2 The generalizability of strategies

Our point-based interactive editing method, coupled with the proposed edge-localized point selector strategy, exhibits transferability to other editing tasks. This is primarily because the editing strategy is intrinsically linked to the characteristics of the editing method itself. The underlying intuition and rationale stem from the implementation mechanism of point-based interaction, which is not inherently constrained by the specific editing task. However, the extent of performance improvement may vary across different tasks, depending on the attributes of the objects being edited.

In contrast, the score-based gradient-guided editing method and the zero-latent source region replacement strategy are more closely tailored to the specific requirements of this particular task. These approaches were designed to address distinct features of the editing target and are therefore not suitable for generalized application.

9 Conclusion

In this paper, motivated by advances in editing techniques, we seek to extend their application to broader scenarios. To this end, we introduce an editing task focused on eye manipulation. However, existing datasets are inadequate for our target domain. We therefore first constructed a dedicated dataset and trained models adapted to the new data distribution. We then fine-tuned these models and evaluated them using existing editing methods, which revealed a limitation in handling edits involving large angular changes. To address these issues, we proposed two distinct improvment strategies tailored to different types of editing approaches. Experimental results demonstrate that our methods significantly improve editing performance, thereby extending the applicability of existing techniques to this specific task and filling a critical gap in this area of research.

Abbreviations

EMDD, eye multi-direction dataset; GAN, generative adversarial network; LPIPS, learned perceptual image patch similarity; MSE, mean squared error; SSIM, structural similarity index measure; VAE, variational autoencoder.

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Author contributions

All authors contributed to the study conception and design. Material preparation, data collection, and analysis were performed by YF, JD, and RW. The first draft of the manuscript was written by YF, RW, and QG. All authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Data availability
The VideoAttnTarget datasets analyzed in this study are available in the Columbia Gaze dataset repository at https://www.cs.columbia.edu/CAVE/databases/columbia_gaze/. The datasets generated and analyzed during the current study are available at https://github.com/function-y-x/benchmark-Drag-and-beyond.

Declarations

Competing interests
Ming-Ming Cheng and Qing Guo are Associate Editors at Visual Intelligence and were not involved in the editorial review of this article or the decision to publish it. The authors declare that they have no other competing interests.

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